

The Islamia University of Bahawalpur

Faculty of computer science

Department of (IT)



(Project Proposal)

WEB-BASED FAKE NEWS DETECTION STSTEM

Submitted By:

Muhammad Umar

S25BINFT7M04026

Submitted To:

Sir Munawar Ahmad

Semester:

7(M_1) ADP

March , 2026

Project Proposal

WEB-BASED FAKE NEWS DETECTION STSTEM

1. Project Title

Fake News Detection Website

2. Project Description

The Fake News Detection System is a web-based platform designed to identify and analyze news articles to determine whether they are real or fake. The main objective of this project is to develop a user-friendly website that allows users to enter news content or links and receive an analysis of its credibility.

The system will use data analysis and machine learning techniques to evaluate news articles based on patterns commonly found in fake or misleading information. Users will be able to submit news text or URLs, and the system will process the information and provide results indicating whether the news is likely to be real or fake.

The platform will also include an administrative dashboard where administrators can manage datasets, review submitted news, and monitor system performance. This will help improve the accuracy and reliability of the system over time.

The project aims to provide a simple and efficient tool that helps users verify the authenticity of online news and reduce the spread of misinformation. It will also demonstrate practical knowledge of web development, database design, and basic machine learning integration.

3. Problem Statement

With the rapid growth of social media and online news platforms, the spread of fake news has become a major problem worldwide. False information can mislead the public, create confusion, and negatively influence public opinion.

Many users find it difficult to determine whether the news they read online is trustworthy or misleading. Currently, there are limited tools available that allow users to quickly verify the authenticity of news content.

Therefore, there is a need for an automated system that can analyze news content and help users detect whether it is real or fake. A web-based fake news detection system can help reduce misinformation and improve awareness about reliable news sources.

4. Objectives of the Project

The main objectives of this project are:

- Develop a web-based platform for detecting fake news.
- Allow users to submit news articles or text for analysis.
- Provide automatic classification of news as real or fake.
- Provide an admin panel to manage news datasets and system data.
- Improve public awareness about misinformation.
- Provide quick and reliable results to users.
- Ensure responsive design for mobile and desktop devices.

5. Scope of the System

The system will include the following functionalities:

- User Side
 - User registration and login
 - Submit news text or URL for analysis
 - View detection results (Real or Fake)
 - View previously analyzed news
 - View credibility score of the news
- Admin Side
 - Admin login
 - Manage news datasets
 - Monitor submitted news
 - Update or train detection models
 - Manage users

6. System Features

The proposed system will provide several important features including:

User Authentication:

Users must register and log in to use the fake news detection service.

News Analysis:

Users can submit news articles or links to check whether the news is real or fake.

Machine Learning Detection:

The system will analyze text using machine learning models trained on news datasets.

Admin Management:

Administrators can manage datasets, monitor system usage, and improve model performance.

Search Functionality:

Users can search previously analyzed news articles.

User-Friendly Interface:

The system will provide a simple and responsive interface for easy use.

7. Technologies to be Used

❖ Front-End Technologies

- HTML
- CSS
- JavaScript
- Tailwind css (for responsive design)
- React
- Axios

❖ Back-End Technologies

- Node.js
- Express js
- REST APIs
- Json

❖ Database

- MySQL

❖ Development Tools

- Visual Studio Code
- Git for version control
- NPM/Yarn
- GitHub

8. Database Design (Main Tables)

Users Table

- UserID
- Name
- Email
- Password

News Table

- NewsID
- Title
- Content
- Source
- SubmittedBy

Analysis Table

- AnalysisID
- NewsID
- Result (Real/Fake)
- ConfidenceScore
- Date

Admin Table

- AdminID
- Name
- Email
- Password

9. Methodology

The system will be developed using the Software Development Life Cycle (SDLC).

Requirement Analysis:

Understanding the system requirements and user needs.

System Design:

Designing the database structure, system architecture, and user interface.

Implementation:

Developing the system using web technologies and machine learning models.

Testing:

Testing the system to ensure accurate detection and proper functionality.

Deployment:

Launching the system on a web server.

Maintenance:

Improving the system and updating datasets over time.

10. Expected Outcomes

The system is expected to:

- Help users identify fake news easily.
- Reduce the spread of misinformation online.
- Provide fast and reliable analysis of news content.
- Improve public awareness about news credibility.
- Demonstrate the use of web development and machine learning technologies.

11. Conclusion

The Web-Based Fake News Detection System will help users verify the authenticity of news articles and reduce the spread of misinformation. By combining web technologies with machine learning techniques, the system will provide an efficient tool for detecting fake news. This project will also demonstrate practical applications of data analysis, web development, and database management in solving real-world problems.